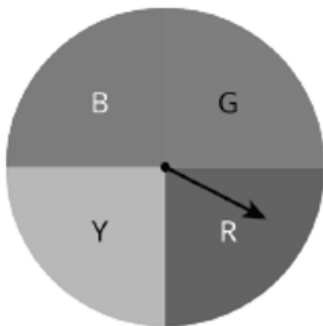


1. Explain the difference between theoretical and experimental probability of a simple experiment.

Theoretical is $\frac{\text{outcome}}{\text{total events in the sample space}}$

Experimental is $\frac{\text{times the event occurs}}{\text{total number of trials}}$

Jennifer performed a simulation and spun the arrow on the spinner below. She recorded his results in the frequency table below.



Color	Frequency
Blue	8
Green	3
Red	7
Yellow	4

2. What is the theoretical probability of landing on green?

$$\frac{1}{4}$$

3. What is the experimental probability of landing on green?

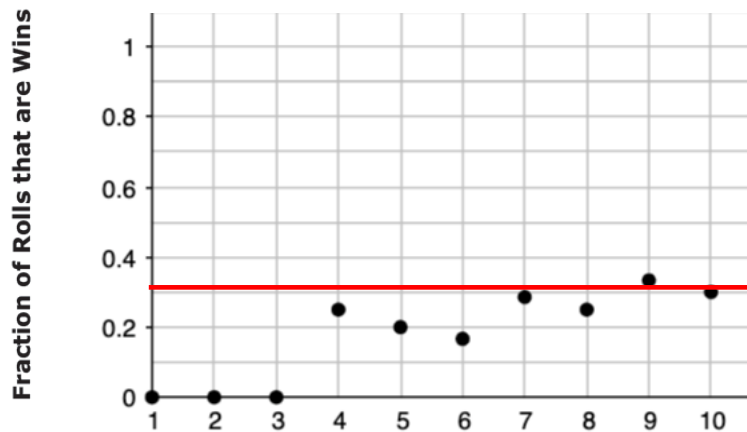
$$\frac{3}{8+3+7+4} = \frac{3}{22}$$

Tyler is playing a game where he wins if he rolls a 2 or a 5 on a standard number cube.

4. What is the theoretical probability that Tyler wins the game?

$$\frac{2}{6} \text{ or } \frac{1}{3}$$

He then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.

0.3

6. What type of probability is this?

Experimental

7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 2 or a 5.

8. After 10 rolls, what fraction of the total rolls were a win?

$\frac{3}{10}$

9. If Tyler rolled the number cube 10 more times, estimate the probability that he will roll a 2 or a 5.

$\frac{1}{3}$

10. If he were to roll the number cube 500, how many times would you expect him to roll a 2 or a 5.

$\frac{1}{3}(500) \approx 167$ times